

Module - I: Allotropes of Carbon

The different form of the same element having different physical properties but almost similar chemical properties. The phenomena of existence of allotropic form of an element called allotropy.

Graphite

In graphite, each carbon atom sp^2 hybridized and it is covalently bonded by another three carbon atom and one π bond (one -ve charge) are free.

Graphite consists of 2D hexagonal rings in layer. The layers are arranged in difference of 340 pm. Due to large difference between these layers indicates that only weak ~~van~~der-waals forces holds these layers together. Due to this, graphite become soft in nature, slippery nature. That's why graphite used as lubricant.

Graphite is a good conductor of heat and electricity due to free electron layer by layer.

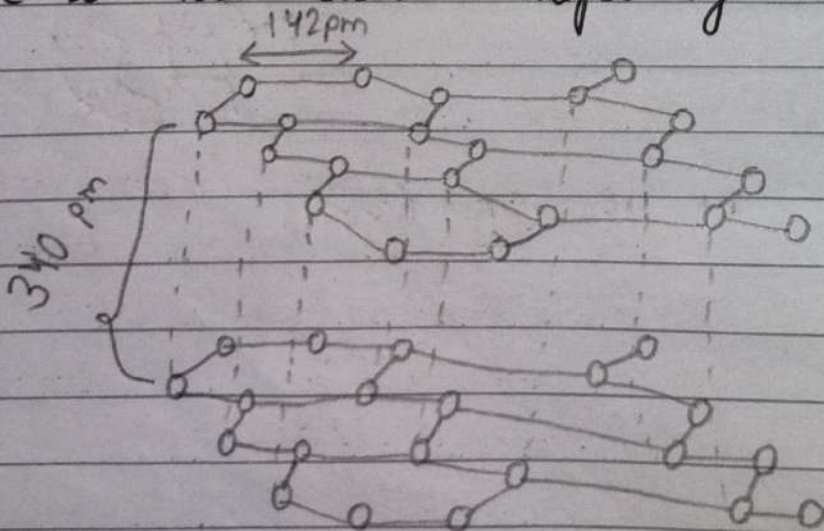
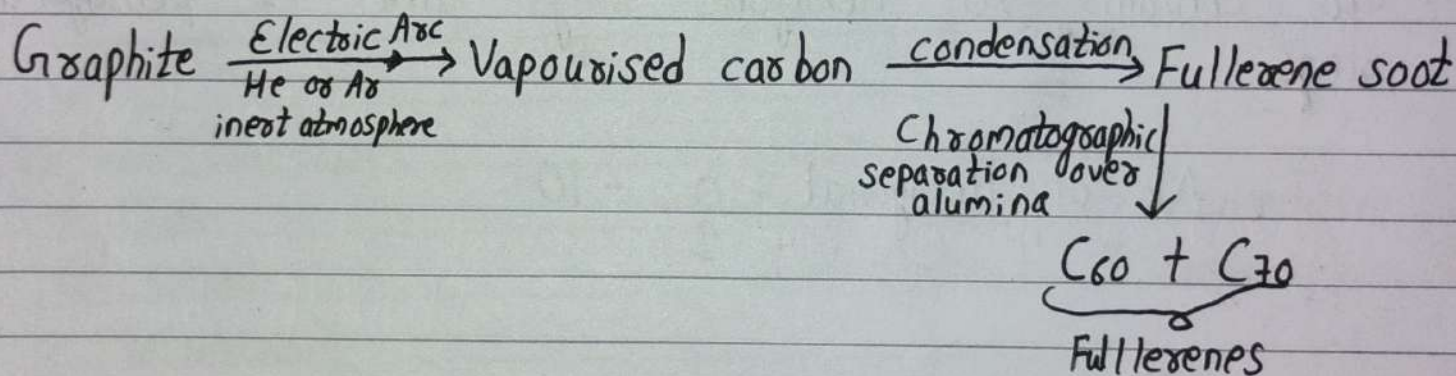


Fig: Structure of Graphite.

Fullerene

- In 1985, scientist discovered a new family of carbon allotrope, consisting of cluster of carbon atoms such as C_{32} , C_{50} , C_{60} , C_{70} , C_{84} , etc. These are called fullerenes.
- These were discovered by H.W. Kroto, R.F. Curl and R.E. Smalley.
- In 1996, awarded by the nobel prize for their discovery of C_{60} , called as buck-minister-fullerene. It was the name of an American architect.



- Fullerenes are made by heating graphite in electric arc in the presence of inert atmosphere of He or Ar. The material formed by the condensation of vapourised small molecules consist of mainly C_{60} with a small amount of C_{70} .
- C_{60} and C_{70} obtained by the extraction with aromatic hydro-carbon solvent (benzene toluene, etc.)

Q) In which atmosphere, the fullerene is prepared by vapourizing graphite rod?

Ans $\Rightarrow$ In inert atmosphere of Helium / Argon.

$\rightarrow$ Individual members can be separated by chromatographic separation over the alumina.

$\rightarrow$ The purified fullerenes have very attractive colour
i.e. C_{60} in toluene purple and C_{70} gives orange.

$\rightarrow$ Carbon of fullerenes sp^2 hybridised.

$\rightarrow$ C_{60} contains 20 hexagonal rings and 12 pentagonal rings.

$$\text{No. of Hexagonal} = \frac{n}{2} - 10$$

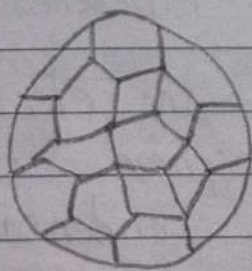


Fig: Structure of Fullerene